

# MINIMUM LAP TIME OPTIMIZATION

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ABSTRACT.

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## 1. THE PROBLEM SETUP

Here we need to explain:

- (1) What we want to do / optimize?
- (2) Which data we know ?
- (3) What assumptions we use ?

The objective of this project is to compute a trajectory that allows a race car to complete a given race track in minimum time. Unlike path-following problems, where the trajectory is prescribed in advance, both the trajectory and the control inputs are optimized while satisfying the physical limitations of the vehicle.

The optimization is carried out on a racing track located in Freiburg. The image of the track is processed to extract the boundaries, from which the centerline is obtained. Using this centerline, we compute the curvature profile and determine the track width along the circuit. These quantities provide a mathematical description of the track and serve as the geometric input for the optimization problem.

Besides the track geometry, the relevant vehicle parameters, such as the wheelbase, admissible steering angle, acceleration limits and maximum lateral acceleration, are assumed to be known. The objective is therefore to determine the vehicle states and control inputs that minimize the total lap time while ensuring that the vehicle remains within the track boundaries and satisfy all imposed constraints.

The next step is to choose an appropriate mathematical model describing the vehicle dynamics. To this end, several simplifying assumptions are introduced in the following section.

## 2. VEHICLE MODEL

In order to formulate our optimization problem, we first need to decide on some simplifications. The goal of this project is not to focus on the vehicular dynamics of a race car, but rather on the optimization problem itself.

Therefore, we decided to use the commonly used kinematic bicycle model to describe the dynamics of the car.

### 2.1. The bicycle model.

The commonly used kinematic bicycle model is a simplification of the dynamics of a car. It assumes that the car has two wheels, one in the front and one in the back. Only the front wheel is used for steering and the slip angle is neglected.

The model is described by the position of the car in global coordinates  $(x, y)$ , the velocity  $v$ , the heading  $\psi$  and the steering angle  $\delta$ . The dynamics of the model are given as

$$\begin{aligned}\dot{x} &= v \cos(\psi) \\ \dot{y} &= v \sin(\psi) \\ \dot{\psi} &= \frac{v}{L} \tan(\delta) \\ \dot{v} &= a.\end{aligned}$$

The constants  $L$  and  $a$  denote the wheelbase of the car, i.e. the distance between the front and rear wheel, and the acceleration of the car respectively. For more details on this model, see [Raj12].

However, for driving around a given track it is inconvenient to use global coordinates because it is more complicate to formulate the constraints. Additionally, we only need the position on the track and do not care for the absolute position in global coordinates.

Therefore, we want to formulate the model using track coordinates. The centerline distance traveled will be given as  $s$  and  $e_y$  denotes the distance from the centerline,  $e_\psi$  denotes the angle between the cars heading and the tangent of the centerline and  $v$  still denotes the velocity of the vehicle.

Now, we can split the velocity into the forward component along the track tangent given as  $v \cos(e_\psi)$  and the sideways component normal to the track tangent given as  $v \sin(e_\psi)$ , which directly leads to  $\dot{e}_y = v \sin(e_\psi)$  because the lateral offset changes exactly by the sideways component of the velocity.

On a curved track, we need to take into account that a car with lateral offset to the centerline needs to travel a different distance compared to a car being on the centerline in order to achieve the same progress. This leads to the relation **EXPLAIN!**

$$\dot{s} = \frac{v \cos(e_\psi)}{(1 - \kappa(s)e_y)},$$

with  $\kappa(s)$  being the curvature of the track at position  $s$ .

Now, for the heading deviation, we calculate

$$e_\psi = \psi - \psi_c(s)$$

and differentiating yields

$$\dot{e}_\psi = \dot{\psi} - \dot{\psi}_c(s)\dot{s}.$$

The cars own heading with respect to the tangent of the centerline changes because of steering and the heading of the track changes because of the curvature of the track. Therefore, we get

$$\dot{e}_\psi = \frac{v}{L} \tan(\delta) - \kappa(s)\dot{s}.$$

This yields the following model

$$z(t) = \begin{bmatrix} s(t) \\ e_y(t) \\ e_\psi(t) \\ v(t) \end{bmatrix} \quad \dot{z}(t) = \begin{bmatrix} \dot{s}(t) \\ v(t) \sin(e_\psi(t)) \\ \frac{v}{L} \tan(\delta(t)) - \kappa(s(t))\dot{s}(t) \\ a(t) \end{bmatrix}$$

EXPLAIN LIMITATIONS OF THE MODEL

### 3. OPTIMIZATION PROBLEM

#### 3.1. Formulation of the optimization problem.

We need to expand the section explaining carefully:

- (1) Decision variables
- (2) Objective
- (3) Constraints
- (4) Boundary conditions

In order to formulate the optimization problem, we need to have some controls of the car. These are given as

$$u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix}$$

with  $a$  being the acceleration and  $\delta$  being the steering angle. With this, we can finally formulate the optimization problem. Using the model above, we can formulate the problem as follows

$$(3.1) \quad \begin{array}{ll} \text{minimize} & T \\ & z(\cdot), u(\cdot), T \end{array}$$

$$(3.2) \quad \text{subject to} \quad \dot{z}(t) = \begin{bmatrix} \dot{s}(t) \\ v(t) \sin(e_\psi(t)) \\ \frac{v}{L} \tan(\delta(t)) - \kappa(s(t)) \dot{s}(t) \\ a(t) \end{bmatrix} =: f(z(t), u(t)), \quad t \in [0, T],$$

$$(3.3) \quad -\frac{w(s(t)) - b}{2} \leq e_y(t) \leq \frac{w(s(t)) - b}{2},$$

$$(3.4) \quad -a_{\min} \leq a(t) \leq a_{\max},$$

$$(3.5) \quad -\delta_{\max} \leq \delta(t) \leq \delta_{\max},$$

$$(3.6) \quad 0 \leq v(t) \leq v_{\max},$$

$$(3.7) \quad \left| \frac{v(t)^2}{L} \tan(\delta(t)) \right| \leq a_{y,\max},$$

$$(3.8) \quad s(0) = 0, \quad s(T) = S,$$

$$(3.9) \quad v(0) = 0, \quad e_y(0) = 0, \quad e_\psi(0) = 0.$$

The constants used in the definition of the optimization problem are defined as follows:

$w(s)$ : track width at the longitudinal position  $s$ ,

$b$ : width of the vehicle,

$L$ : wheelbase of the vehicle,

$a_{\min}$ : magnitude of the maximum permissible deceleration,

$a_{\max}$ : maximum permissible acceleration,

$\delta_{\max}$ : maximum absolute steering angle,

$v_{\max}$ : maximum permissible velocity,

$a_{y,\max}$ : maximum permissible lateral acceleration,

$S$ : total length of the track.

#### 4. NUMERICAL SOLUTION

We need to cover here:

- (1) Discretization
- (2) Multiple-shooting
- (3) CasAdi
- (4) IPOPT (or other solver use)

## REFERENCES

- [Raj12] Rajesh Rajamani. *Vehicle Dynamics and Control*. 2nd ed. Springer US, 2012. ISBN: 9781461414339. DOI: [10.1007/978-1-4614-1433-9](https://doi.org/10.1007/978-1-4614-1433-9).